

Accountable Anonymity for Public Money (RETIRED DRAFTING SKELETON)

Mandatory Bilateral Non-Deniability for Unlinkable Bearer Payments, Without a Trusted Opener

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Public money is going programmable, and programmability invites surveillance: every proposed central-bank digital currency couples the convenience of digital cash to a ledger its issuer can read, freeze, and mine. Physical cash resists this but offers the opposite extreme – no accountability at all, no issuance trail, no recourse. This paper presents BUCK Notes, a bearer-instrument layer for an identity-aware stablecoin that occupies neither pole. *Bulk* traceability is defeated – no analytics firm or omnibus subpoena can reconstruct who paid whom – while *bilateral* disclosure is preserved: the two parties to any payment always retain the cryptographic means to prove, under compulsion and with **no trusted third party**, exactly who the counterparty was. The load-bearing property is the **non-deniable-receipt invariant**: there is no path, even for a *colluding* payer and payee, to construct a payment for which no court-compellable receipt naming the KYC-bound counterparty exists. We formalise the invariant, give a construction meeting it for public-issuer notes (shipped and evaluated on Ethereum), prove its security, and characterise precisely the boundary at which the guarantee stops being free: when *both* parties are anonymous at their respective binding moments, a collusion-resistant receipt provably requires in-circuit set-membership over the identity registry. We give the negative result that the obvious cheaper fix is vacuous, and an incentive-compatible mechanism that bounds the residual risk to the always-identifiable recipient.

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1 Introduction

FC-first framing. Open on the public-money mission; the cryptography serves it.

The choice a society makes about the privacy of its money is a choice about the balance of power between the individual and the institutions that watch over them.

1.1 Contributions

- **The non-deniable-receipt invariant** (Def. 2): a security definition for payment systems whose operative, novel clause forbids a *colluding* payer/payee pair from constructing a payment with no accepting receipt. The accountability is **bilateral** and held by the counterparties; there is no system-wide opening authority.
- **A construction meeting it for unlinkable bearer payments** (public-issuer notes, A1/B1), defeating bulk on-chain traceability while preserving court-compellable bilateral disclosure – **shipped and evaluated** on Ethereum.
- **A forced taxonomy** (Thm. 4): the realisable design space is exactly three flavours; a private-issuer bearer note is structurally impossible.
- **A characterisation of where accountability is free vs. expensive** (Sec. 7): the receipt is free exactly when at least one party is named at the binding instant; when both are anonymous, we show the obvious cheap coupling is *vacuous* (Lem. 11) and that a collusion-resistant receipt requires registry set-membership ZK (Prop. 15).
- **An incentive-compatible backstop** (Prop. 16): for the doubly-anonymous case, an economic mechanism bounds the residual risk to the always-KYC-identifiable recipient, obviating the expensive cryptographic fix under a stated threat model.
- **A reproducible artifact**: every theorem ships an executable `py_ecc` witness on BN254; the contracts, circuits, and verifiers are open and test-covered.

2 Related Work

We place BUCK Notes in the lineage of digital cash and privacy-preserving payment systems, and isolate the property no predecessor provides: a *mandatory, bilateral, collusion-resistant* receipt with *no system-wide opener*.

System	Unlinkable	Mandatory receipt	Collusion-resistant	Trusted opener?	Bearer/paper
Chaum e-cash ¹	issuer-blind	no	no	central bank	no
Zerocash / Zcash ²	yes	no (opt. viewkey)	no	no	no
Tornado / Privacy Pools ³	yes	no (innocence)	no	no	no
PRCash ⁴	yes	regulator-facing	via auditor	yes (auditor)	no
UTT ⁵	yes	budget-gated	via committee	yes (committee)	no
PEReDi ⁶	yes	regulator-facing	via threshold	yes (maintainers)	no
Platypus ⁷	yes	regulator-facing	via central bank	yes (cen. bank)	no
CBDC (e-CNY, digital euro) ⁸	no	n/a	n/a	issuer sees all	no
BUCK Notes (A1/B1)	yes	yes	yes	no	yes
BUCK Notes (A2, reserved)	yes	yes (burden)	partial (Sec. 7)	no	yes

The auditable-privacy line (PRCash, UTT, PEReDi, Platypus) is closest in spirit and the right comparison for a financial-cryptography audience: all four restore accountability to unlinkable payments, but all four vest a *designated authority* – an auditor, a committee, the central bank – with the power to de-anonymise. BUCK’s accountability is *counterparty-held*: the only authority is the per-identity KYC issuer, subpoenaed one named party at a time, never breadth-first across the system.

¹Chaum, D. "Blind Signatures for Untraceable Payments." CRYPTO 1982. [TODO exact cite]

²Ben-Sasson, E., et al. "Zerocash: Decentralized Anonymous Payments from Bitcoin." IEEE S&P 2014. [TODO]

³Buterin, V., Schaefer, J., Tromer, E., et al. "Blockchain Privacy and Regulatory Compliance: Towards a Practical Equilibrium (Privacy Pools)." 2023. [TODO]

⁴Wuest, K., Kostiaenen, K., Capkun, V., Capkun, S. "PRCash: Fast, Private and Regulated Transactions for Digital Currencies." FC 2019. [TODO verify authors/title]

⁵Tomescu, A., et al. "UTT: Decentralized Ecash with Accountable Privacy." IEEE S&P 2022. [TODO verify]

⁶Kiayias, A., Kohlweiss, M., Sarencheh, A. "PEReDi: Privacy-Enhanced, Regulated and Distributed Central Bank Digital Currencies." ACM CCS 2022. [TODO verify]

⁷Wuest, K., Kostiaenen, K., Delius, N., Capkun, S. "Platypus: A Central Bank Digital Currency with Unlinkable Transactions and Privacy-Preserving Regulation." ACM CCS 2022. [TODO verify]

⁸Bank for International Settlements. "Rise of the Central Bank Digital Currencies." [TODO]

3 Model, Adversaries, and the Invariant

An *Identity* is a KYC-certified point $M \in \mathbb{G}_1$ bound to a registered account by an issuer’s Pointcheval-Sanders credential and stored ElGamal-encrypted as $E_{\text{addr}}[X] = (R_X, C_X) = (r_X G, M_X + r_X pk_X)$ with $pk_X = sk_X G$. A *payment* moves value v from payer P to recipient D .

3.1 Adversaries

Definition 1. *Three properties, held against adversaries of increasing power:*

- (**Bulk-tracing resistance.**) *Mallory, a passive observer with full chain history and unbounded compute but no party’s secrets, cannot reconstruct who paid whom, how much, or link a mint to its spend(s).*
- (**Third-party non-disclosure.**) *Nothing on chain reveals either party’s Identity point M to a non-participant, even in aggregate across many payments – including to the issuer (below).*
- (**Mutual decryptability / accountability.**) *Each of the two participants, from data they hold plus public chain state, can produce a sound receipt naming the other’s registered M ; and a colluding pair cannot arrange a payment for which no such receipt exists.*

The first two properties must hold against the strongest realistic deanonymiser, the **Issuer**: the KYC trust anchor that holds, for every client it certified, the plaintext identity (id_i, m_i, M_i) and the credential σ_i it issued – strictly more than Mallory knows. We treat a compelled or wholly breached issuer in Sec. 3.3.

3.2 The non-deniable-receipt invariant

Definition 2. (Non-deniable receipt.) *A protocol satisfies the **non-deniable-receipt invariant** if for every accepted payment there exist a receipt $\mathcal{R}$ and an efficient public predicate RcptVerify such that:*

- (**Completeness.**) *The recipient D can compute $\mathcal{R}$ from data they hold plus public chain state, and $\text{RcptVerify}(\mathcal{R}) = 1$ names a registered Identity M_P and value v .*
- (**Soundness / non-frameability.**) *No PPT adversary produces $\mathcal{R}'$ with $\text{RcptVerify}(\mathcal{R}') = 1$ naming an Identity that did not author the payment.*
- (**No deniability.**) *No PPT payer can cause an accepted payment to D for which no accepting receipt naming the payer exists – even in collusion with D .*

The third clause is the operative one and the departure from prior art: it is the collusion-resistant form. Completeness alone protects a self-interested recipient; the invariant additionally forbids a *willing* pair from constructing an untraceable payment.

3.3 Issuer-blindness: accountability without a bulk opener

The KYC issuer is the system’s only trust anchor, and the natural objection to any *accountable privacy* claim is that the issuer is a bulk-deanonymisation backdoor. It is not. Issuance is *address-blind*: the issuer signs the identity scalar $m = H(id)$ once (a Pointcheval-Sanders credential over m alone) and is thereafter done; the client derives each on-chain account *offline* – a fresh key pair (sk, pk) , a self-formed ElGamal credential $E_{\text{addr}} = (rG, M + rpk)$, and a *rerandomised* credential

σ' – without the issuer ever seeing pk . The issuer’s view is therefore, per certified client i , the tuple $(id_i, m_i, M_i, \sigma_i)$ – the plaintext identity – plus public chain state; it holds *no* chain addresses.

Theorem 3. (Issuer-blindness.) *Under statistical unlinkability of PS rerandomisation (Thm. 2 of the companion Proofs) and IND-CPA security of ElGamal on $\mathbb{G}_1$ (equivalently DDH), an issuer – even one compelled or wholly breached, in possession of $\{(id_i, m_i, M_i, \sigma_i)\}_i$ and the entire chain – has negligible advantage in matching any issued identity M_i to its on-chain registration (pk_j, E_j, σ'_j) . Compulsion or breach yields the set of certified identities but no map to chain accounts; deanonymisation remains per-account, requiring that account’s sk or a counterparty receipt.*

Proof. Two locks. *Credential lock.* σ'_j is a rerandomisation of some issued σ_i ; by statistical unlinkability the two carry zero mutual information, so even an unbounded issuer cannot recognise its own signature. *Plaintext lock.* The issuer *knows* M_i , so its task is not decryption but *plaintext-checking*: does $E_j = (R, C)$ encrypt M_i ? That is deciding whether $(G, pk_j, R, C - M_i)$ is a DDH tuple, and an algorithm answering it for a chosen M_i breaks IND-CPA (submit $m_0 = M_i$, m_1 random). Knowing the plaintext thus confers no advantage. With both locks shut and no stored pk , the issuer is reduced to an independent DDH decision per (identity, account) pair. $\square$

This is the load-bearing form of *no trusted opener*: the closest thing to an authority cannot open transactions in bulk, even against itself – under compulsion it can only *verify* a link some party surrenders (map a receipt-disclosed M to the real person), never *produce* one. Two caveats bound the theorem, and we state rather than bury them:

- **Computational, not information-theoretic.** The plaintext lock is DDH; only the credential lock is statistical. A discrete-log break – in particular a quantum computer running Shor against $\mathbb{G}_1$ – turns the harvested $\{M_i\}$ plus the archived chain into a wholesale, retroactive deanonymisation of all *past* identity and A2-Note traffic. The breached or compelled issuer’s KYC records are precisely that harvest corpus (companion *Proofs*, Implementation Notes: *Harvest-Now, Decrypt-Later*).
- **Retroactive, not prospective.** Address-blind issuance protects *past* registrations unconditionally against *future* issuer compromise; but a coerced issuer can change the protocol going forward (demand pk at KYC, record addresses, backdoor credentials). Issuer-blindness is a statement about issuances already performed under the honest protocol.

4 Construction: A Note Is a Deferred Approve Handshake

A standard transfer between two registered identities is a synchronous bilateral **approve** handshake: each party re-encrypts their credential under the other's key and proves (Chaum-Pedersen) the re-encryption carries the same M . A Note reorganises the same handshake through a Tornado-style commitment pool, splitting it across time so that mint and spend are unlinkable, while preserving the bilateral decryptability.

4.1 The pool, the commitment, the nullifier

4.2 The receipt as a four-link proof chain

4.3 Three flavours: public-issuer (A1/B1) now, private-issuer (A2) reserved

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5 The Forced Taxonomy

Who can be named, and how, is fixed by what is known at mint time – the axis that defines the flavours. One of the four nominal cells is not merely undesirable but impossible.

Theorem 4. (No private-issuer bearer note.) *A meaningful bearer note whose issuer Identity is hidden from third parties cannot exist.*

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6 Security of the Public-Issuer Construction (A1, B1)

This is the complete result: for public-issuer notes the non-deniable-receipt invariant holds in all three clauses, with no open problem. The issuer is bound at *mint* by a Schnorr signature over the batch commitment under their registered key; the depositor's reverse leg is the `verifyApprove`-shaped re-encryption at *deposit*.

Theorem 5. (Issuer-binding completeness.) *An honestly generated public-issuer Schnorr proof verifies.*

Theorem 6. (Issuer-binding soundness.) *If `verifyIssuerSchnorr` accepts, the batch commitments were authorised by the issuer's registered key. Reduction: Schnorr EUF-CMA / special soundness.*

Theorem 7. (Receipt completeness.) *For every note accepted by `mint` and redeemed by `spend` $\neq$ `spendACP`, the recipient computes $\mathcal{R}$ with `RcptVerify`($\mathcal{R}$) = 1 naming a registered Identity. Links (a),(b),(d) hold in the shipped contracts; (c) holds by issuer-binding soundness.*

Theorem 8. (Receipt soundness / non-frameability.) *No PPT adversary produces an accepting $\mathcal{R}'$ naming an Identity that did not author the payment. A forged public receipt breaks Schnorr EUF-CMA or the collision-resistance of the commitment hash.*

Theorem 9. (No deniability under collusion – public issuer.) *No payer/recipient coalition can produce an accepted public-issuer payment with no accepting receipt: `mint` rejects any leaf lacking a valid issuer binding, so every spendable leaf carries one, recoverable by the holder of the opening.*

Corollary 10. (Mutual decryptability, public issuer.) *Combining the issuer half (Schnorr binding at `mint`) with the depositor half (`verifyDepositorForIssuer` at *deposit*), both parties recover the other's registered Identity and each can prove it under compulsion – with no third party.*

7 The Cost of Anonymity: the Private-Issuer Boundary (A2)

The public-issuer result is free of any SNARK over identities because, at the binding instant, one party is *named*. We now show that removing the last named anchor – making *both* parties anonymous at their respective binding moments (the A2 setting: issuer named only at mint, recipient named only at spend, mint and spend unlinkable) – is exactly what makes collusion-resistance expensive, and we locate the boundary precisely.

7.1 The EOA-transfer mirror: the cost is the anonymity

A direct private-to-private transfer achieves mutual decryptability with no SNARK and free collusion-resistance, because the mandatory Chaum-Pedersen-bound **approve** reads *both* parties' registered records in a single call – both anchors co-present, both named. A2 splits the anchors in time by anonymity: the issuer's anchor exists only at mint, the recipient's only at spend. No on-chain check reads both at once, so the A2 binding *blinds* the recipient's key to bridge them – and blinding it is precisely what drops the "under the recipient's *registered* key" guarantee.

7.2 The obvious fix is vacuous * NEW (rigorous, ready to formalise) *

Lemma 11. (Vacuity of key-equality coupling.) *Let an A2 mint constrain only that the issuer payload $eIss$ and the note ciphertext E_{note} are encrypted under the same key pk_{rec} , with the recipient's identity point M_{rec} a free witness. Then for any issuer-chosen pk_{bogus} the constraint is satisfiable while the note remains spendable by the intended recipient D and $eIss$ is encrypted under $pk_{bogus} \neq pk_D$.*

Proof. Set $pk_{rec} := pk_{bogus}$ and $M_{rec} := M_D + r_n(pk_D - pk_{bogus})$. Then $E_{note} = (r_n G, M_{rec} + r_n pk_{rec}) = (r_n G, M_D + r_n pk_D)$, a valid encryption of M_D under D 's registered key, so **verifySpendCP** accepts and D spends; while $eIss = (r' G, M_{iss} + r' pk_{bogus})$ is under pk_{bogus} , so a compelled sk_D decrypts $eIss$ to $M_{iss} + r'(pk_{bogus} - pk_D) \neq M_{iss}$. The free witness M_{rec} absorbs the constraint. $\square$

7.3 Closing the clause: the coupled construction

Lemma 11 shows that binding $eIss$ to a key is worthless; the binding must be to the recipient's *registered* key, and the issuer – who does not hold sk_D – can only assert that key *exists in the registry*. This yields the coupled mint circuit $\mathcal{C}^*$. Write the registry as an accumulator with public root rt over leaves $\ell_X = \text{Poseidon}(\text{addr}_X, pk_X, E_{\text{addr}}[X])$; the issuer is $I = \text{msg.sender}$ with public $E_{\text{addr}}[I] = (R_I, C_I)$; the note ciphertext is $E_{note} = (R_n, C_n)$ and the issuer payload $eIss = (R_e, C_e)$ is a public output (the leaf-tie of Sec. 4). Per A2 leaf, with private witnesses $(sk_I, M_{iss}, pk_D, E_{\text{addr}}[D] = (R_D, C_D), \pi_D, r', r_n)$, $\mathcal{C}^*$ constrains:

- (C1, issuer opening) $pk_I = sk_I G, \quad M_{iss} = C_I - sk_I R_I;$
- (C2, membership) $\text{MerkleVerify}(rt, \pi_D, \text{Poseidon}(\cdot, pk_D, E_{\text{addr}}[D])) = 1;$
- (C3, re-randomisation) $R_n = R_D + r_n G, \quad C_n = C_D + r_n pk_D;$
- (C4, payload coupling) $R_e = r' G, \quad C_e = M_{iss} + r' pk_D.$

(C1) ties $eIss$ to the issuer's *registered* identity (the public $E_{\text{addr}}[I]$); (C2) certifies the recipient is a *real* registered account without revealing which (leaf private, root public); (C3) makes E_{note} a re-randomisation of that account's registered credential – hence an encryption of M_D under pk_D ;

(C4) encrypts $eIss$ under the *same* pk_D . The spender’s key is pinned to pk_D only later, at spend, by `verifySpendCP`; soundness is therefore a *composition* of the mint membership proof and the spend equality proof – the only two moments at which the issuer’s and the recipient’s anchors are respectively live.

The construction rests on one property of registration, already present in the identity layer for unrelated (determinism / unlinkability) reasons, and now load-bearing.

Definition 12. (Non-grindable identities.) *Each registered identity point is $M = H(\text{id}) \cdot G$, where id is the KYC-verified attribute record and H (keccak256 over a canonical encoding) is preimage-resistant, modelled as a random oracle. No PPT adversary registers an identity point of its choosing; in particular it cannot register an account whose M equals a prescribed target $M_D + \delta G$ for an adversarially chosen $\delta \neq 0$.*

Proposition 13. (Sufficiency.) *Under Assumption 12, with `verifySpendCP` enforced at spend, the coupled construction $\mathcal{C}^*$ closes the no-deniability clause (Def. 2) for A2: for every accepted mint-and-spend, a compelled spender recovers the issuer’s registered M_{iss} from $eIss$.*

Proof. By (C3), E_{note} is a re-randomisation of $E_{\text{addr}}[D]$, so for every sk , $C_n - sk R_n = M_D + (r_D + r_n)(sk_D - sk)G$, where $sk_D = \text{dlog } pk_D$. At spend, `verifySpendCP` establishes that the spender holds sk_s with registered key $pk_s = sk_s G$ and that $M_s := C_n - sk_s R_n$ equals the spender’s registered identity. Substituting, $M_s = M_D + (r_D + r_n)(sk_D - sk_s)G$. If $sk_s \neq sk_D$, then $M_s = M_D + \delta G$ with $\delta = (r_D + r_n)(sk_D - sk_s) \neq 0$; by Assumption 12 the probability that such a point is a registered identity distinct from D ’s is negligible (the adversary would have to register an account at $M_D + \delta G$, i.e. find KYC data hashing to $m_D + \delta$, contradicting preimage resistance). Hence $sk_s = sk_D$ and $pk_s = pk_D$ except with negligible probability. By (C4), $C_e - sk_s R_e = M_{iss} + r'(pk_D - pk_s) = M_{iss}$, and by (C1) M_{iss} is the issuer’s registered identity. The compelled spender therefore produces a receipt naming I . $\square$

Proposition 14. (Recipient privacy preserved.) *In $\mathcal{C}^*$ the values $pk_D, E_{\text{addr}}[D], M_{iss}$ are private witnesses; the only public artefacts are the fixed root rt and the ciphertext $eIss$ (semantically secure absent sk_D). By Groth16 zero-knowledge and ElGamal semantic security, no third party learns the recipient’s identity or key, and two A2 mints to distinct recipients are computationally indistinguishable.*

This sufficiency is what reserves A2 rather than abandons it: the construction exists and is sound, but its non-native $\mathbb{G}_1$ arithmetic (C1, C3, C4 are emulated $\mathbb{F}_p$ operations on the proof system’s own curve) costs $\sim 10^6$ R1CS per leaf (Sec. 8) – an order of magnitude beyond the public-issuer path, and the reason A2 ships under the mechanism of Sec. 7.5 in the interim.

Remark (the cost is prover-side; interactive A2). The 10^6 R1CS is a *prover* cost: the EVM verifies the resulting proof in constant gas, so the construction is expensive to *prove* but cheap to *check*. Where the wallet runs on a free-compute substrate that already holds the registry tree (here, a Holochain layer), the prover cost is absorbed off chain and the tree need not live on chain beyond its root. A variant we note for completeness: when both parties are co-present off chain at note creation (*interactive A2*), they run the full bilateral handshake and prove it at mint – recovering the EOA transfer’s free, collusion-resistant mutual decryptability (Sec. 7) while still depositing unlinkably – at the cost of an online recipient. Neither observation relaxes Prop. 15 (the membership proof is still required); they relocate *where* it is proved and confirm its on-chain verification was never the bottleneck.

7.4 Is the SNARK necessary? * NEW (conditional result + open problem) *

The sufficiency above is an upper bound. The matching question – *must* a collusion-resistant A2 receipt use registry set-membership? – is the paper’s central open problem. We resolve it under one restriction and leave the unconditional statement open.

Proposition 15. (Conditional necessity.) *Let Π be a protocol that, in the A2 setting, (i) preserves bulk-tracing resistance and third-party non-disclosure (issuer named only at mint, recipient only at spend, mint and spend unlinkable), (ii) closes the no-deniability clause, and (iii) achieves the binding by referencing a registered party at a single protocol step. Then Π performs a zero-knowledge proof of membership in the identity registry – over the recipient set at mint, or the issuer set at spend.*

Proof. By Lemma 11, (ii) forces Π to certify that $eIss$ decrypts, under the addressed recipient’s registered key, to a registered *issuer* identity – binding to some key alone does not close the clause. This certificate references a registered party: the recipient’s key pk_D (at mint), or the issuer’s identity (at spend). By (iii) it is a single step’s reference, either in the clear or in zero knowledge.

In the clear is ruled out. Revealing the referenced key or identity on chain names a party and makes them linkable across mint and spend, violating (i): naming pk_D at mint exposes the recipient, and revealing the decrypted M_{iss} at spend exposes the issuer. Importing E_{note} (or a deterministic commitment to it) to the mint to bind there instead re-links mint and spend, also violating (i).

Zero-knowledge is set-membership. A reference that certifies the referenced party is registered without revealing *which* – the recipient’s pk_D at mint, or the issuer that $eIss$ re-encrypts at spend – is, by definition, a zero-knowledge proof of membership in the registry. As (i) excludes the in-clear case, Π performs this. $\square$

The restriction (iii) – a single binding step – is what the proof leans on, and relaxing it names the open frontier. A binding amortised across steps, a two-party mint in which the recipient contributes a blinded witness without exposing E_{note} , a witness-encryption construction, or an accountability mechanism that does not bind $eIss$ at all (threshold escrow of M_{iss} to a regulator key – a weaker trust model) each escape the case analysis. Whether any achieves collusion-resistant A2 naming without registry set-membership is open (Sec. 9); a clean impossibility would prove the SNARK necessary, a loophole would be a cheaper design.

7.5 A mechanism instead of a proof * NEW (model to formalise) *

Proposition 16. (Recipient-burden incentive compatibility.) *Model an A2 deposit by recipient D : audit probability q , AML penalty L for holding funds from an unaccountable source, delivery-verification cost ε (decrypt $eIss$, run the γ -coupling), and collusion benefit b . If $qL > b$ and ε is small, the unique sequential equilibrium is "verify at delivery, reject an un-nameable note", so accepted A2 notes are nameable with high probability. By non-frameability the only off-equilibrium outcome is an un-nameable note (never a framed third party), and D – always KYC-identifiable – bears the penalty; the residual risk is therefore self-limiting to the recipient.*

8 Implementation and Evaluation

8.1 What is shipped

8.2 Costs

8.3 Verification methodology

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9 Discussion, Limitations, and Future Work

9.1 The autonomy thesis, restated

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10 Appendix A: Executable Theorem Witnesses

Each theorem ships a compact `py_ecc` sanity check on BN254 that runs in CI alongside the wallet's vector generation; the same vectors drive the circom witness tests and the Solidity verifier tests, keeping the reference math, the circuit, and the on-chain verifier in lock-step.

```
# Representative witness (stub): Lemma 'vacuous' -- the bogus-eIss attack.
# Asserts E_note decrypts to M_D under pk_D (spendable) while eIss is
# under pk_bogus (compelled sk_D -> garbage). Full witnesses live in
# alberta_buck/wallet + test/vectors/.
M_rec = M_D + r_n * (pk_D - pk_bogus)          # free witness absorbs the constraint
E_note = (r_n * G, M_rec + r_n * pk_bogus)     # == (r_n*G, M_D + r_n*pk_D)
assert elgamal_decrypt(E_note, sk_D) == M_D     # note is spendable by D
eIss = (r_p * G, M_iss + r_p * pk_bogus)       # bound under bogus key
assert elgamal_decrypt(eIss, sk_D) != M_iss     # D cannot name the issuer
```

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